REPORT 005: EPOCH 5: INTERACTIVE RUST EGUI CONTROLLER SURFACE, VECTOR KNOBS WITH SOFT-TAKEOVER GHOST RINGS & MAPPING INSPECTOR

2026-11-04 00:30:00 | Epoch 5 of 20: src/ui_editor.rs (Multi-Device Header Bar, Korg nanoKONTROL2 & Akai MIDImix vector surfaces, Soft-Takeover ghost rings, and Inspector Side Panel with Nord Drum 2 assignment)

Operator: Gemini 4 Argon (Medium) · Follows: report-004

Prompt

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Epoch 5 of 20: Implement the interactive Rust egui controller surface and visual
↪  editor in src/ui_editor.rs styled with yaae-theme and twoBowls aesthetics (crisp
↪  zero-rounding borders, high-contrast tabular readouts, AccentPalette highlights).
↪  Render: (1) a Multi-Device Header Bar showing all connected and remembered
↪  devices with USB hardware fingerprint badges, connection status indicators, quick
↪  theme/background selector, and virtual output status; (2) purpose-built visual
↪  controller layouts mirroring the Korg nanoKONTROL2 (transport deck + 8 track
↪  columns with custom vector rotary knobs showing target value + physical
↪  soft-takeover ghost ring, vertical sliders, and LED-lit S/M/R buttons) and Akai
↪  MIDImix (3 knob rows + fader + M/S/R buttons + master strip); and (3) an
↪  Inspector Side Panel for editing any selected control's CC/Note number, channel,
↪  label, curve, min/max range, button behavior, and Nord Drum 2 target parameter
↪  assignment. Add headless egui/state tests (with CARGO_BUILD_JOBS=2), git commit
↪  all changes, and compile reports/report-005.html from reports/report-005.md via
↪  /home/renick/claude/html-report-generator/build-report.sh --pdf and commit.
```

TL;DR

Implemented `src/ui_editor.rs` (`ControllerEditorUiState`, `ValueDisplayFormat`, `render_multi_device_header_bar`, `render_nanokontrol2_surface`, `render_akai_midimix_surface`, `render_inspector_side_panel`, `draw_vector_rotary_knob`, `draw_vector_vertical_slider`, `draw_led_hardware_button`, and `render_controller_workspace`). Renders the Multi-Device Header Bar with USB `HardwareFingerprint` badges, quick `yaae-theme` (15 themes) and `twoBowls` background (4 presets) selectors, purpose-built vector layouts for the Korg nanoKONTROL2 and Akai MIDImix featuring Soft Takeover ghost rings and real-time delta indicators, and the Inspector Side Panel with one-click Nord Drum 2 Desktop CC assignment across all 32 synthesis/FX parameters. Verified by 36 passing automated tests (`30` Rust unit/headless `egui` tests + `6` Node.js SDK tests).

1. Plain-Language Overview & Epoch 5 Goals

When configuring or performing with a physical MIDI controller, a musician needs a visual mirror of their hardware on screen—showing every knob, fader, and button in its exact physical arrangement. More importantly, when Soft Takeover locks a knob after a preset change (because the physical knob is sitting at `25` while the new preset's target is `90`), the musician needs to see **both** positions at once: the active software value (`90`) and a **ghost ring** showing where the physical knob is (`25`), along with an arrow (`^+65` or `v-65`) indicating which way to turn the knob to unlock it.

In **Epoch 5 of 20**, we built `src/ui_editor.rs`, the interactive Rust `egui` controller surface and visual

inspector styled with `yaae-theme` and `twoBowls` aesthetics (`egui::Rounding::ZERO` crisp borders, high-contrast monospace tabular readouts, and `AccentPalette` status highlights):

1. **Multi-Device Header Bar (`render_multi_device_header_bar`)**: Displays tabs for all connected and remembered hardware controllers with their USB `HardwareFingerprint` badge (`[0944:0117::nanoKONTROL2]`), `ONLINE`/`OFFLINE` status, ALSA port address (`24:0`), quick `SoftTakeoverMode` (`Catch`/`Pickup`/`Jump`) and `LedOperatingMode` (`Int`/`Ext`/`8opPerc`) toggles, dropdowns for all 15 `yaae-theme` palettes and 4 `twoBowls` background presets, `DEC`/`HEX` readout toggle (`ValueDisplayFormat`), and virtual output port status (`V-OUT: nkEditor4 Out [ON]`).
2. **Purpose-Built Vector Controller Surfaces (`render_nanokontrol2_surface` & `render_akai_midimix_surf`**
   (a) **Korg `nanoKONTROL2`**: Left 11-button Transport Deck (`Cycle`, `Marker Set/< />`, `Track < />`, `Rewind`, `Forward`, `Stop`, `Play`, `Record`) + 8 Track columns, each with a custom vector rotary knob (`draw_vector_rotary_knob`), LED-lit `S` / `M` / `R` buttons (`draw_led_hardware_button`), and a custom vertical vector slider (`draw_vector_vertical_slider`).
   (b) **Akai `MIDI Mix`**: 8 Track columns with 3 rotary knob rows (`knob1`, `knob2`, `knob3`), `MUTE` / `SOLO` / `REC` Note-LED buttons, and a track `Fader`, plus the right `MASTER` strip (`Bank <`, `Bank >`, `Solo Master`, `Master Fader`).
3. **Inspector Side Panel (`render_inspector_side_panel`)**: Allows editing any clicked control's output message type (`CC` vs `Note`), output number (`0..=127`), MIDI channel (`1..=16`), custom label, polarity inversion, `ResponseCurve` (`Linear`, `Exponential`, `Logarithmic`, `SCurve`), `min_value`/`max_value` range, `Toggle`/`Momentary` button behavior, `on_value`/`off_value`, and one-click **Nord Drum 2 Desktop** parameter assignment (`assign_nord_drum_2_parameter`) across all 32 synthesis and Global FX CCs.

Plain-English Concepts Mapped to Code Components

Plain-English Concept	What It Does for the Musician	Exact Code Component
Multi-Device Header Bar	Shows hardware fingerprint badges, ALSA port status, and quick theme/takeover/LED selectors	`src/ui_editor.rs` (`render_multi_device_header_bar`, `ControllerEditorUiState`)
Vector Rotary Knob + Ghost Ring	Draws the 270-degree target sweep arc, Soft Takeover physical ghost ring, and `^`/`v` delta readout	`src/ui_editor.rs` (`draw_vector_rotary_knob`, `draw_arc`)
Vector Vertical Slider + Ghost Bar	Draws the vertical fader level, Soft Takeover physical ghost line, and tabular CC/value readout	`src/ui_editor.rs` (`draw_vector_vertical_slider`)
Hardware LED Button Widget	Renders crisp zero-rounding buttons with `Solo` (amber), `Mute` (green), and `Rec` (red) LED bars	`src/ui_editor.rs` (`draw_led_hardware_button`)
Korg nanoKONTROL2 & Akai MIDImix Decks	Mirrors the exact physical layout of all 51 nanoKONTROL2 controls and 60 MIDImix controls	`src/ui_editor.rs` (`render_nanokontrol2_surface`, `render_akai_midimix_surface`)
Control Inspector & Nord Drum 2 Assigner	Edits CC/Note mapping parameters and assigns any of the 32 Nord Drum 2 synthesis/FX CCs in one click	`src/ui_editor.rs` (`render_inspector_side_panel`, `ControllerEditorUiState::assign_nord_drum_2_parameter`)

2. Phase 1: Custom Vector Primitives & Purpose-Built Controller Layouts (‘src/ui_editor.rs‘)

Crisp Zero-Rounding Vector Widgets and Soft Takeover Ghost Visualization

All custom widgets in `src/ui_editor.rs` use `egui::Painter` with `Rounding::ZERO` and `ThemeSettings::resolved_palette()` (`AccentPalette`):

- **`draw_vector_rotary_knob`**: Allocates an interactive click-and-drag cell, draws a 270-degree arc (`-225 deg` to `+45 deg`) for the active target value in `palette.accent`, and—whenever `SoftTakeoverStatus` is present—draws an outer ghost marker and arc segment between `physical_value` and `target_value` in `palette.solo_active_bg` while locked, displaying the live directional delta (`^-65` or `v+15`) in the monospace readout until the knob catches the target.
- **`draw_vector_vertical_slider`**: Renders a crisp vertical fader track, filled target bar, fader cap in `palette.accent_strong`, and a horizontal physical ghost bar across the fader trench whenever Soft Takeover is locked.
- **`draw_led_hardware_button`**: Renders hardware buttons with a top LED indicator bar (`palette.solo_active_bg` for `S`, `palette.mute_active_bg` for `M`, `palette.record_arm_active_bg` for `R/REC`, and `palette.accent` for transport LEDs). Clicking any button selects it in the Inspector and feeds a press event through `DeviceMappingPipeline` and `DeviceLedController`.

3. Phase 2: Inspector Side Panel & Nord Drum 2 Target Assignment

Live Mapping Mutation and Synthesizer Target Binding

`render_inspector_side_panel` binds directly to `ui_state.selected_control_path` inside the active `DeviceMappingPipeline`:

- Edits `output_message_type` (`Cc` or `Note`), `output_number` (`0..=127`), `output_channel` (`1..=16`), `label`, `invert`, continuous `min_value/max_value` and `ResponseCurve`, or button `behavior` (`Toggle/Momentary`) and `on_value/off_value`, immediately calling `pipeline.update_entry(entry)`.
- **`ControllerEditorUiState::assign_nord_drum_2_parameter`**: Selects any pad (`Pad 1..6` -> MIDI Channels `1..6`) and any of the 32 `nord_drum_2::ALL_PARAMS` (`Tone`, `Noise`, `Click`, `Mix`, or `GlobalFx`). Per-pad parameters automatically route to the selected pad channel (`1..=6` with label `"ND2 P4 TIMBRE"`), while `NordDrum2Section::GlobalFx` parameters (`DIST_AMOUNT`, `EQ_GAIN`, `ECHO_AMOUNT`, etc.) automatically route to `nord_drum_2::channels::GLOBAL_FX` (Channel `10` with label `"ND2 FX ECHO_AMOUNT"`).

Tests

- `CARGO_BUILD_JOBS=2 cargo test` — 30 Rust unit and integration tests passing (`2` new headless `egui` and Inspector state test suites in `ui_editor::tests` verifying multi-frame rendering of both Korg `nanoKONTROL2` and Akai `MIDI Mix` surfaces, Soft Takeover ghost ring rendering, `DEC/HEX` formatting, and Nord Drum 2 Pad/Global FX parameter assignment), 0 failures
- `npm test` (`node --test js/*.test.js`) — 6 Node.js SDK unit tests passing, 0 failures

QA

`36 tests passing (30 Rust + 6 Node.js)`

`Headless egui multi-frame rendering of Korg nanoKONTROL2 (51 controls) & Akai MIDImix (60 controls) verified`

`Soft Takeover physical ghost ring & directional delta indicator verified`

`Inspector Nord Drum 2 Pad (Ch 1..6) & Global FX (Ch 10) assignment verified`

`0 warnings, 0 failures`

All Epoch 5 deliverables compiled cleanly with `CARGO_BUILD_JOBS=2` and passed all Rust and Node.js test suites without launching any blocking GUI windows.

Files touched

- ‘src/ui_editor.rs‘ — Implemented ‘ValueDisplayFormat‘, ‘ControllerEditorUiState‘, ‘render_multi_device_header_bar‘, ‘draw_vector_rotary_knob‘, ‘draw_vector_vertical_slider‘, ‘draw_led_hardware_button‘, ‘render_nanokontrol2_surface‘, ‘render_akai_midimix_surface‘, ‘render_inspector_side_panel‘, and ‘render_controller_workspace‘ + headless ‘egui‘ and state unit tests
- ‘src/lib.rs‘ — Exported ‘pub mod ui_editor‘
- ‘src/main.rs‘ — Connected ‘ControllerEditorUiState‘ and ‘render_controller_workspace‘ into ‘NkEditorApp::update‘

Test coverage (36 total)

QA verdict

PASS

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